

4 (a) A **progressive** wave is a wave in which **energy is carried from one point to another** by means of vibrations or oscillations within the wave. Particles within the wave are **not transported** along the wave. B1

(b) (i) $\lambda = 0.40$ m since distance between P and Q is 1.5 wavelength. C1

$$v = f \lambda = 0.40 \times 850 = 340 \text{ m s}^{-1} \quad \text{A1}$$

(ii)

$$\text{Intensity of sound at man} \quad I = \frac{P}{A} = \frac{1500}{4\pi(80.0)^2} \quad \text{C1}$$

$$= 0.01865 \text{ W m}^{-2} \quad \text{C1}$$

$$\begin{aligned} \text{Power intercepted by man} &= I \times A = 0.01865 \times 2.1 \times 10^{-3} \quad \text{A1} \\ &= 3.9 \times 10^{-5} \text{ W} \end{aligned}$$

5 (c) Destructive interference is when two waves arrive at the same point B1